

Optical Alignment Lab

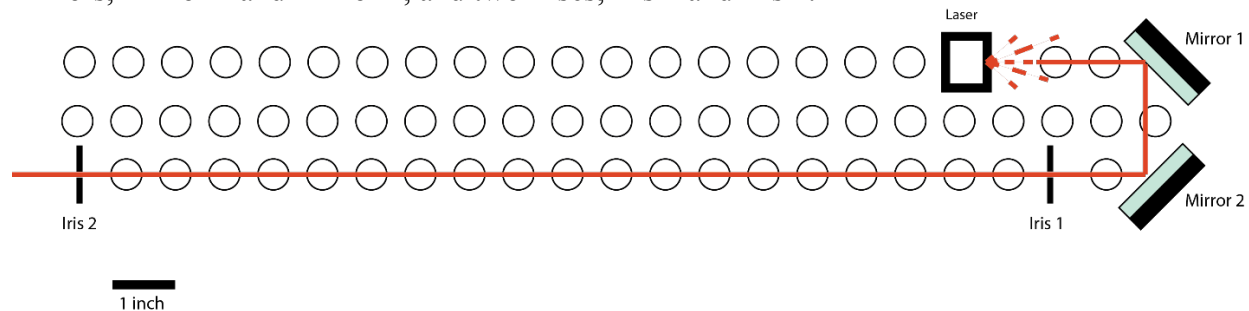
(developed by Prof. Ashley Carter, Amherst College)

The goal for this lab is to learn how to align optical components on an optical breadboard to make an optical system. Typically, optical engineers or laser physicists are only given an optical diagram and are expected to be able to know how to build the optical system from that diagram. This is just like a contractor only getting a blueprint for a house, or an electrical engineer only getting a circuit diagram. Here we will give you several diagrams for various optical systems and have you build those systems. We will start off easy with a simple optical diagram and almost all of the directions to build the optical system. Then, in the last part we will make things more challenging with increasingly more complex optical systems and fewer directions regarding how to build the system.

Part 1 – Align a laser

Goal: Align a laser through two irises.

Optical Diagram: Here is your first optical diagram. Let's see if you can build it! In this diagram, we show you the screw holes on the optical breadboard as a series of circles. You can see the placement of the laser, mirrors, and irises. Essentially, we are going to align the laser beam to the straight line set by the two irises along the screw holes. To do that, you will use two mirrors, Mirror 1 and Mirror 2, and two irises, Iris 1 and Iris 2.



Key design points:

- We want the laser as close to Mirror 1 as possible, to limit the amount the beam will have diverged before it hits Mirror 1.
- We want Mirror 1 and 2 as close to each other as possible so that we will have the maximum range of motion for adjusting the beam.
- We want the irises as far apart from each other as possible, to make the alignment easier and more precise.

Procedure:

1) Snugly mount a laser close to the end of one side of the breadboard, leaving room for the first mirror at the end of the same row. (Make sure to consult the Optics – Rules to Live By! Document for how to choose and tighten screws.)

2) Get two irises out and set both irises so that their holes are a height of 4.5 inches off the optical table. (Again consult the Optics – Rules to Live By! guide for how to screw post holders to bases, how to set an iris hole to a particular height, or how to put a collar on an iris.)

3) Screw down one of the irises where Mirror 2 will be. Or for even better alignment, you can screw down the iris in the same column of screws as where Mirror 1 and Mirror 2 will be, but at the end of the optical breadboard. (Hopefully, by now you are getting the feel for consulting the Optics – Rules to Live By! guide for how to screw things down, or put things together. You'll need to do this again to make sure that you screw down the iris appropriately in the correct position. We'll assume from now on that you will know to consult this guide and will continue on without these notes.)

4) Now we can coarsely align Mirror 1. Do this by placing the mirror on the optical breadboard without screwing the mirror down. Make sure the beam hits the center of the mirror. Then, translate the mirror back and forth along the screw holes just a little bit, getting closer or further away from the laser. (You are moving perpendicular to the iris.) This movement back and forth is one degree of freedom that you have to place the mirror. Notice that by moving the mirror back and forth you change the position of the beam. Now rotate the mirror by loosening the post holder screw and rotating the post attached to the mirror. This is the second degree of freedom that you have to place the mirror. Notice how this movement changes the angle of the beam. Use these two degrees of freedom to center the beam on the iris. Tie the mirror to the optical breadboard using the base clamps.

5) Now screw down both irises in the row next to the laser, as indicated in the optical diagram. One will be close to one end of the breadboard, but will leave room for Mirror 2. The other will be on the other side of the breadboard. (Remember: further apart makes the alignment better).

6) Coarsely align Mirror 2 so that the beam hits the mirror dead on and so that the beam is roughly aligned to the two irises.

7) Now do a fine alignment of the beam to the two irises. To do the fine alignment you are going to use the two knobs on each mirror holder. You'll notice that one knob moves the beam in the x direction (horizontally) and one knob moves the beam in the y direction (vertically). Close Iris 1 and adjust the knobs on Mirror 1 until the beam is aligned to the hole of Iris 1. Then, open Iris 1 and close Iris 2, and adjust the knobs on Mirror 2 so that the beam is aligned to the hole on Iris 2. Repeat this procedure until the beam is aligned to both irises. You'll notice that the beam will slowly move or "walk" into the right position. Thus, this procedure is called "walking the beam". Essentially, what you are doing is setting the position of the beam with the first mirror and the angle of the beam with the second mirror. Now misalign your beam and use that principle to help you walk the beam to the right spot again. Hopefully, it took you a lot less time the second time around. Remember: Mirror 1 controls alignment at Iris 1, and Mirror 2 controls alignment at Iris 2.

Assessment:

1) Intensity check --- One way we could verify that the centering is accurate would be to use a sensor after the second iris and check when maximal light intensity is seen on the sensor - that's when we are best aligned.

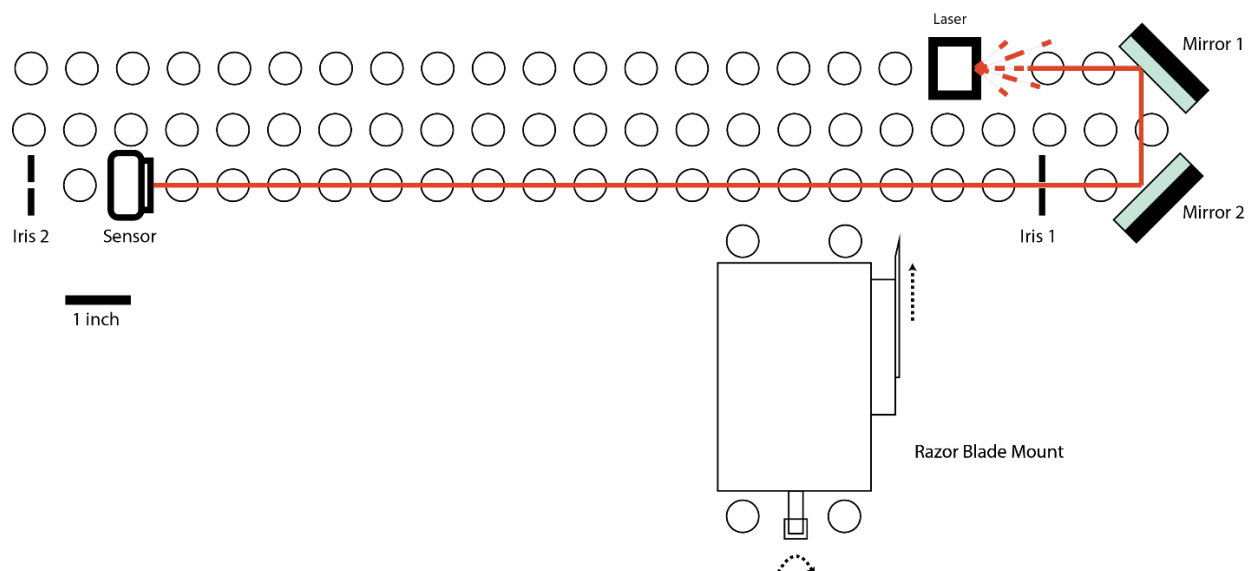
2) Target Practice ---- Or we could place a third iris at a location that is farther from Mirror 2 than Iris 2 and check to see if the beam hits the target.

Part 2 – Measure Beam Width (completely *optional*)

Goal: Measure beam width of our laser.

Design: To measure beam width we will use a power meter and a razor on a micrometer-adjustable mount, with the idea that we will use the razor blade to increasingly block the beam while taking the power readings. The derivative of this data will give you the beam profile. If the beam is Gaussian, you can obtain the width of the beam by fitting a Gaussian to the beam profile. If it is not Gaussian, you can measure the beam width by finding the points where the intensity diminishes by $1/e^2$

Optical Diagram: Here you will build the same setup as in Part 1, expect now you will add a razor blade on a translation stage and sensor (e.g., a photodiode or photoresistor) or a power meter (e.g., a photometer based upon a photodiode or a bolometer). There should be a photometer at every station, but please avoid burning out your detector!



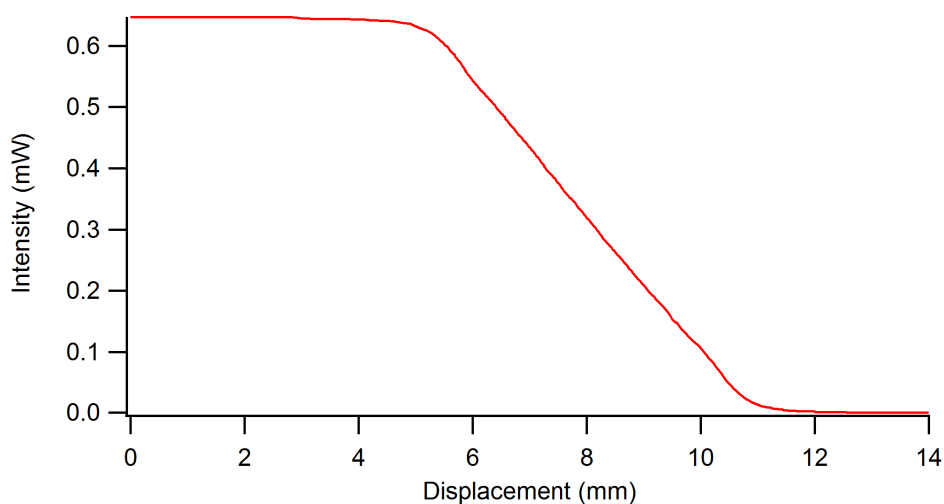
Key Design Points:

- You'll want to place the razor blade so that it is perpendicular to the beam line and translates straight through the beam. You don't want it to cut the beam at an angle.

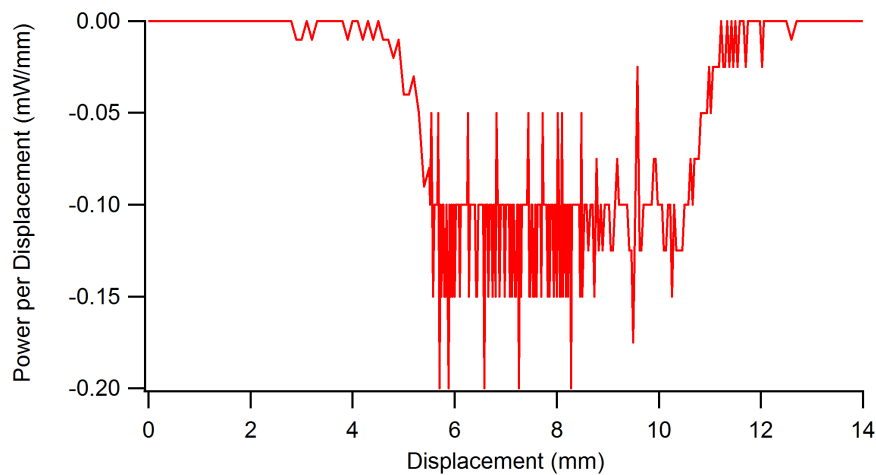
Procedure:

- 1) Install the razor blade mount so that the razor blade translates through the beam after Iris 1. If you place it as close to Iris 1 as possible, this will be helpful for the Assessment for this section. Make sure the range of motion of the razor allows both for full beam to go through and for it to be fully covered.
- 2) Install the sensor/power meter head/receiving optical fiber bundle right before Iris 2.
- 3) The micrometer on the mount should be 20 μm per smallest increment indicated (0.2 mm per rotation) and have a full range of 25 mm. You can check this calibration by extending the micrometer from 0 to max and measuring the extension with a ruler.
- 4) Hook up the power meter head/receiving optical fiber bundle to its measurement system. Look up the user manual on how it operates. Make sure to set the correct settings for wavelength and make sure the maximum allowed power reading is not exceeded (you should be less than 50 mW). If power is greater than max power, then you need to use a neutral density filter.
- 5) Turn off the lights to reduce the background power. Start moving the razor blade through the beam. If you move in 0.1 mm increments, then that should give enough resolution. As you move the razor, record the total power reading on the power meter. Make sure to take enough data points (~ 5) when the beam is not blocked and when the beam is fully blocked.

You might get data like this:



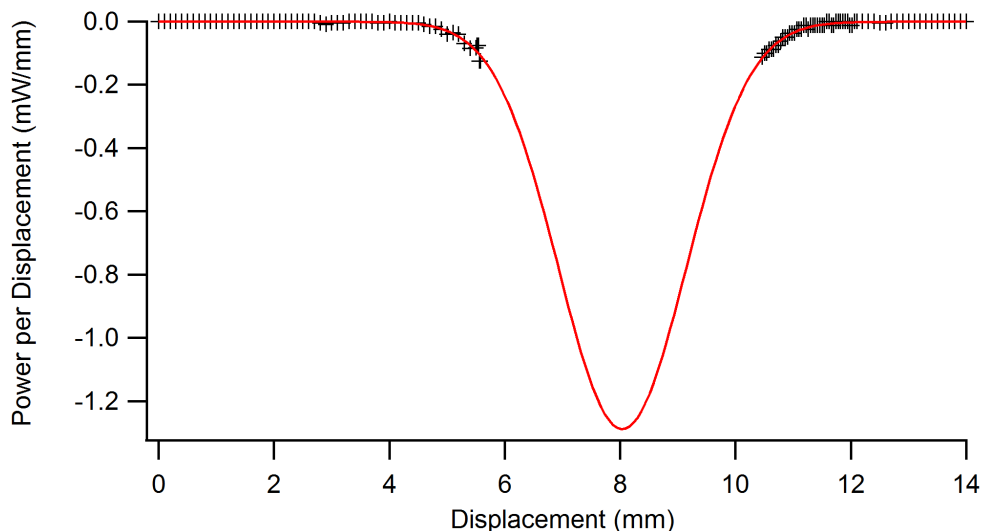
6) Now we took the derivative of this data, simply by calculating $\Delta(y)/\Delta(x)$ between every subsequent point in *IGOR Pro*. Or, you can use the derivative command under the analyze menu. The derivative we obtained looks like the following:



(Negative values arose because of the way we took the data. You can multiply by -1 to get positive values.)

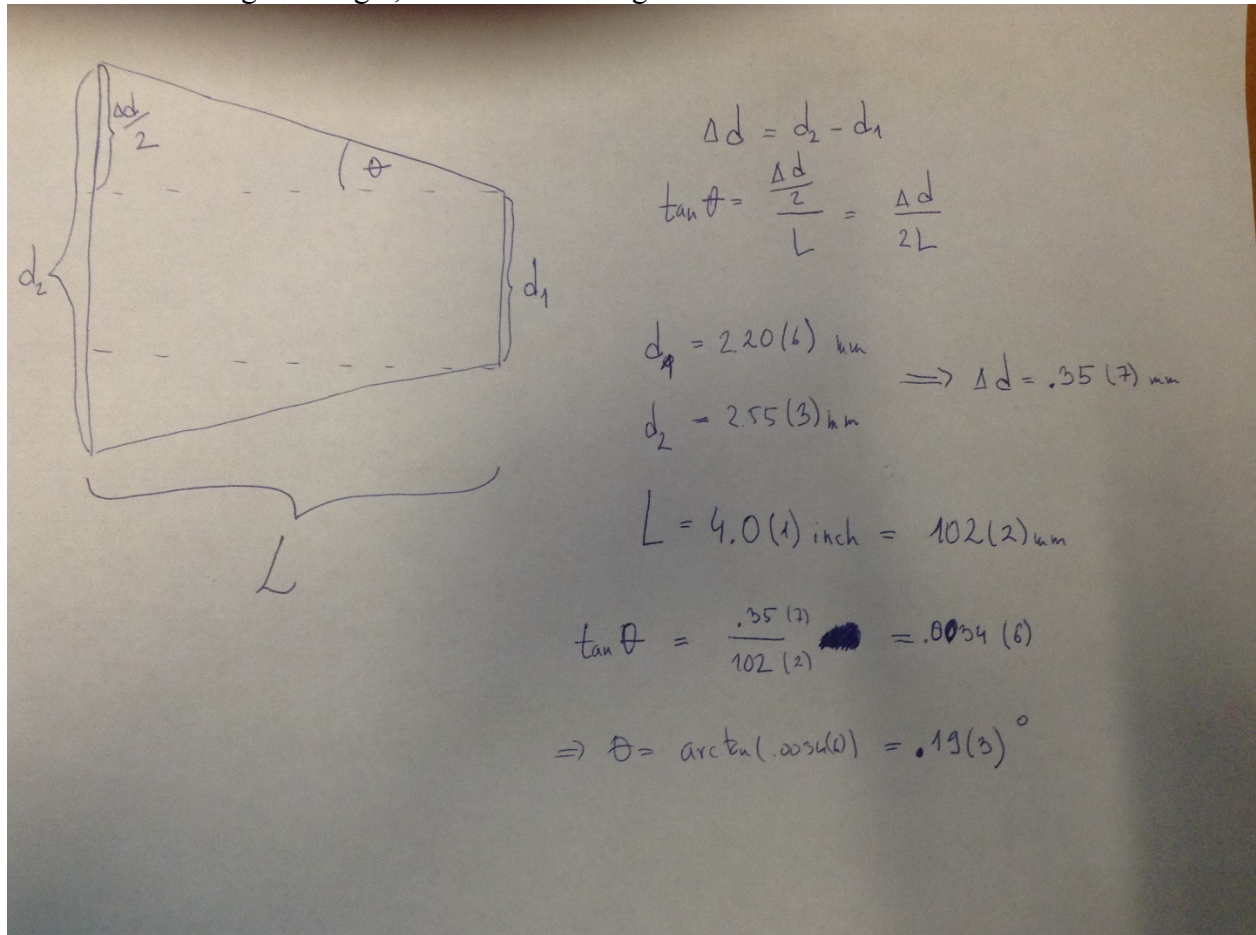
7) Notice that this laser is not Gaussian! It is a cheap diode laser with an elliptical beam. So to calculate the beam width find the maximum ($\text{max} = 0.12 \text{ mW/mm}$) and use that to find the point at which the maximum drops off by $1/e^2$ ($\text{max}/e^2 = 0.016 \text{ mW/mm}$). This point occurs at $\sim 5 \text{ mm}$ and $\sim 11 \text{ mm}$, meaning that the beam width is 6 mm .

8) If the beam *were* Gaussian, we could fit a Gaussian to the beam profile. Here we will just fit the data at the edges of the beam and ignoring the central regions (see figure below). In IGOR the Gaussian fit is of the form: $y_0 + A \exp[-((x-x_0)/(\text{width}))^2]$. So change the width that IGOR reports into the real " $1/e^2$ width" that everyone else uses by using the equation $\text{width} = \text{Igor width}/\sqrt{2}$. Here we find a width of $1.10 \pm .03 \text{ mm}$, which means the full beam diameter is $2.20 \pm .06 \text{ mm}$. Clearly, you can see that this method is intended for Gaussian beams.



Assessment:

1) Measure the beam *divergence* – Measure the beam width at a second location, so we can calculate the divergence angle, θ . Procedure in figure below.



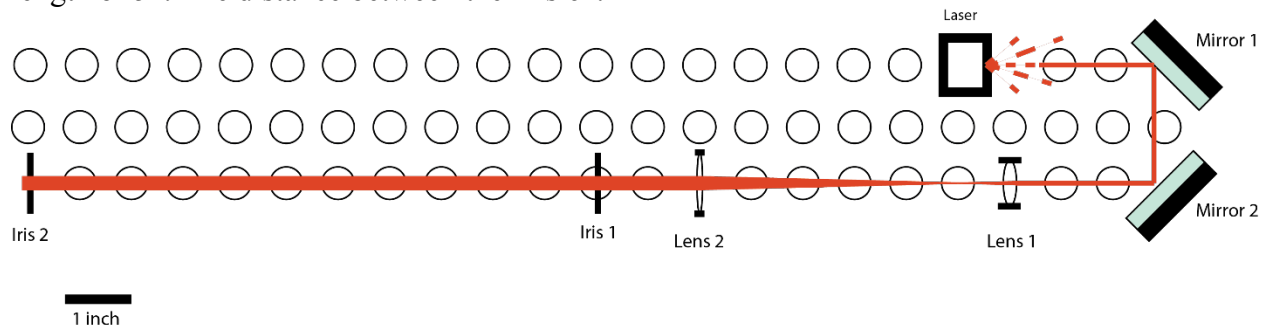
Compare your results for this measurement to those of other groups in the class.

Part 3 – Install a telescope

Goal: Increase the beam width and *collimate* the beam using a telescope.

Design: By using two plano-convex lenses, Lens 1 and then Lens 2, we can build a telescope. Lens 1 has a focal length of f_1 and Lens 2 has a focal length of f_2 . Placing the lenses $f_1 + f_2$ apart allows you to expand or shrink the beam by a factor of f_2/f_1 . Moving one of the lenses so that the distance between the lenses is not quite $f_1 + f_2$ can correct for a diverging or converging incident beam.

Optical Diagram: To expand the beam, the lens with the smaller focal length should be Lens 1, and the lens with the longer focal length should be Lens 2. The distance between Lens 1 and Lens 2 should be equal to $f_1 + f_2$. Here Lens 1 has a focal length of 1" and Lens 2 has a focal length of 5". The distance between them is 6".



Procedure:

- 1) Move Iris 1 to make room for your lenses.
- 2) Make sure your setup is still aligned through the irises with no lenses.
- 3) Select two lenses that you would like to use to expand the beam and mount them in optical mounts.
- 4) Insert Lens 1 close to Mirror 2, but don't bolt it down firmly yet. Let's first coarsely align the lens. To do this, roughly set the height of the post (y) so that the beam hits the center of the lens. Next, make sure that the lens/lens mount is perpendicular to the beam axis by eye.
- 5) Now let's do the fine alignment. First, screw the lens mount to the optical breadboard and adjust the slope of the lens mount to fix any x alignment. The beam should be centered on the lens in x and the beam exiting the lens should be aligned to both irises in x . Next, move the post height (y) more precisely so that the exiting beam is aligned to both irises in y . Tighten the post collar to make sure this height doesn't change. The beam should be centered on the lens in y and the beam exiting the lens should be aligned to both irises in y . Finally, rotate the post attached to the lens until the back reflection of the lens is perfectly on top of the original beam. Tightly bolt down the lens.
- 6) If everything looks good, insert Lens 2 using the same procedure as Lens 1, except with one variation. In the coarse alignment, you'll need to make sure to roughly adjust the z position of the lens. This should be at a distance of $f_1 + f_2$. In the fine alignment, you are going to finely align the z position of the lens. This will set the collimation of the beam. To do this, image the beam very far away (at essentially "infinity") and focus it at that location. Focusing the beam at "infinity" will create a collimated beam. To finely adjust the z position, you can bolt bases to the optical table to create a track in z . Moving the mount back and forth along this track will allow for fine z movements. Translation stages in x , y , or z can be also be used to have control over the fine alignment of the lens.

Assessment:

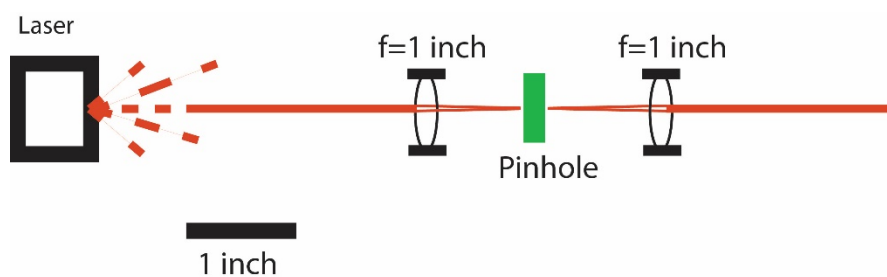
- 1) Measure the beam width --- Check to see if the beam has expanded, by measuring the beam width after Lens 2.
- 2) Measure the beam divergence --- Check to see if the beam is collimated by measuring the beam width in two locations after Lens 2.
- 3) Absent-minded professor – Your instructor could “accidentally” misalign one of your optics. See if you can realign the system *quickly*.

Part 4 – Align an optical system (Optional)

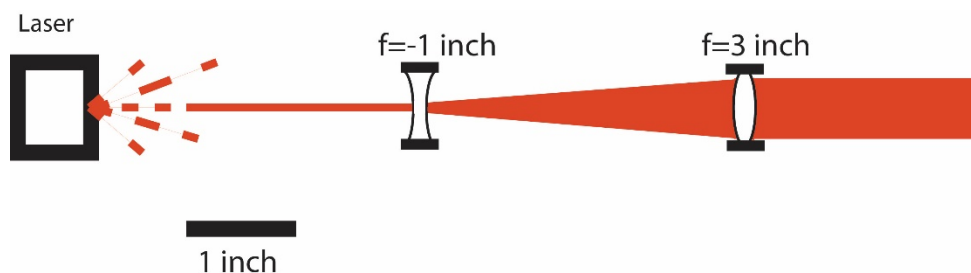
Goal: Build an optical system from an optical diagram. You will build two systems with increasing complexity. Test your skills!

System 1: You can choose either a) the spatial filter or b) the beam expander with the diverging lens.

a)



b)



System 2: You can choose either a) the spectrometer or b) the microscope.

a)

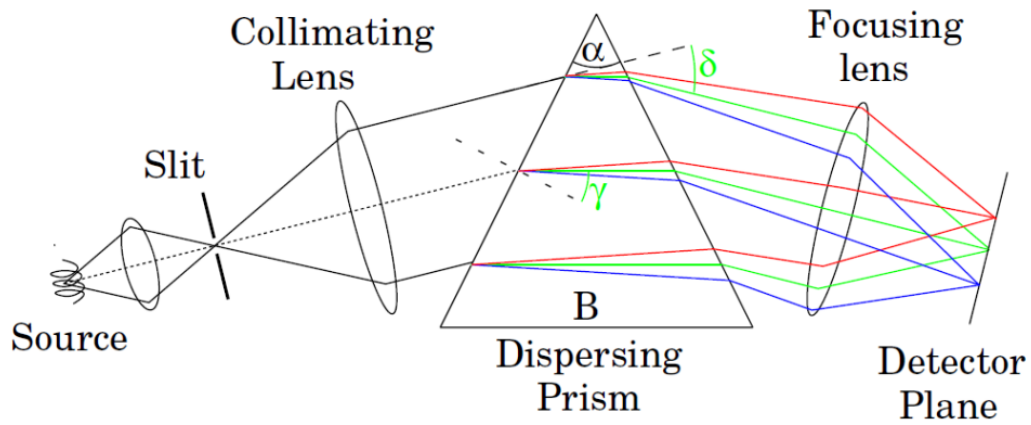


Image Credit: Mcleod Lab – from ECEN Advanced Optics

b)

